Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	25 MARCH 2026
Team ID	69c0f43a1466f700a6d30fcf
Project Name	AUTOMATING EMPLOYEE DATA
Maximum Marks	4 Marks

Technical Architecture (Explanation)

Architecture Type:

3-Tier Architecture (Client–Server Model)

Flow:

1. **User interacts with Web UI (Frontend)**
2. **Request goes to Application Server (Backend Logic)**
3. **Backend processes data and interacts with Database**
4. **Response is sent back to user**

Simple Architecture Diagram (You can draw this)

[User (Browser)]
↓
[Frontend (HTML, CSS, JS)]
↓
[Backend Server (PHP / Node.js)]
↓
[Database (MySQL)]
↓

Guidelines:

- Store employee details in system
- Manage data (Add, Edit, Delete)

The system helps to manage employee data easily, reduces manual work, and improves efficiency."

- User Registration
- Login System
- Add Employee
- Edit Employee
- Delete Employee
- View Employee List

Areas of Project Product Tech Stack

This slide showcases the areas of product tech stack. Its aim is to improve these mentioned areas for a feasible and manageable tech stack to increase efficiency. This slide includes communication, design, project planning, road mapping, HR management, online collaboration, user research, etc.

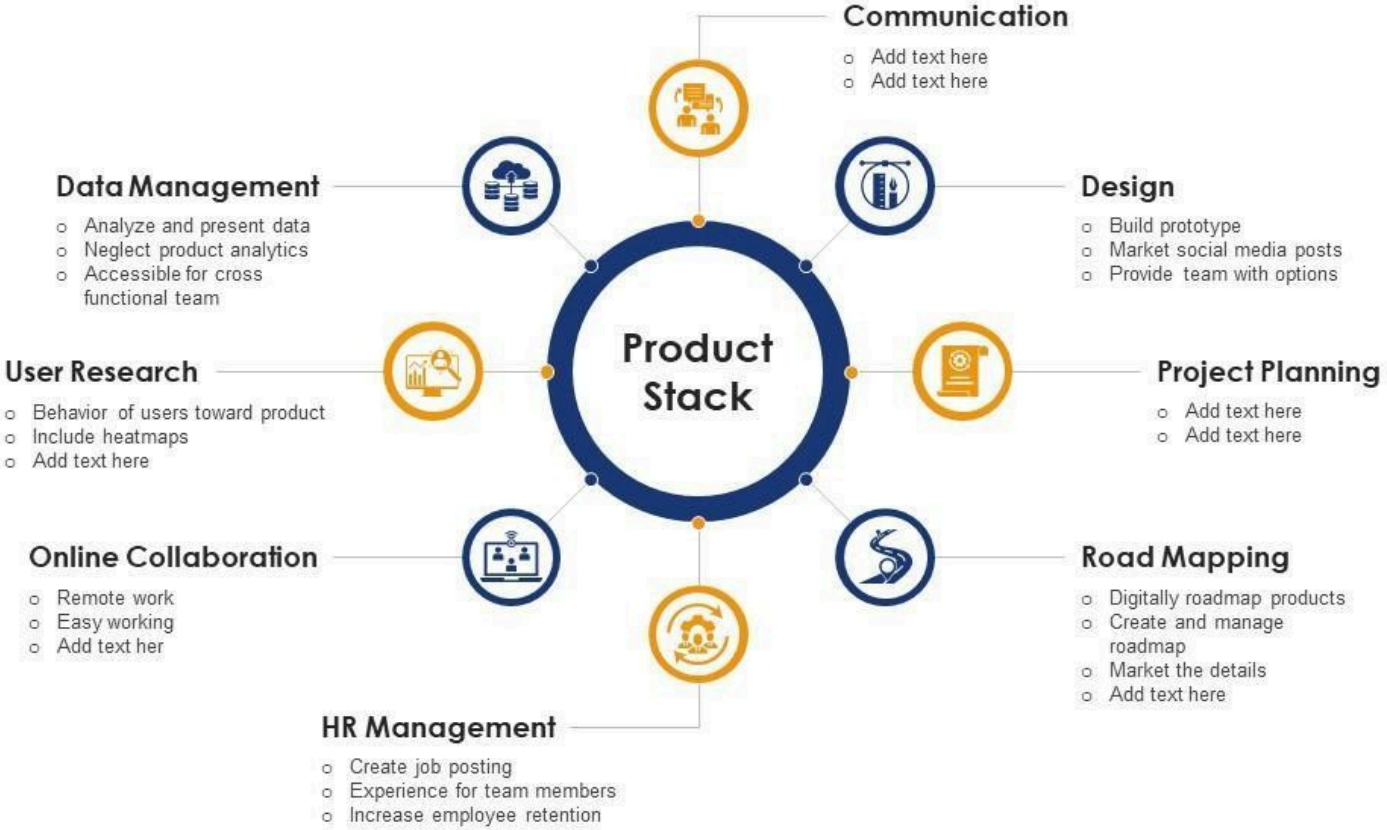

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Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application e.g. Web UI, Mobile App, Chatbot etc.	HTML, CSS, JavaScript / Angular Js / React Js etc.
2.	Application Logic-1	Logic for a process in the application	Java / Python
3.	Application Logic-2	Logic for a process in the application	IBM Watson STT service
4.	Application Logic-3	Logic for a process in the application	IBM Watson Assistant
5.	Database	Data Type, Configurations etc.	MySQL, NoSQL, etc.
6.	Cloud Database	Database Service on Cloud	IBM DB2, IBM Cloudant etc.
7.	File Storage	File storage requirements	IBM Block Storage or Other Storage Service or Local Filesystem
8.	External API-1	Purpose of External API used in the application	IBM Weather API, etc.
9.	External API-2	Purpose of External API used in the application	Aadhar API, etc.
10.	Machine Learning Model	Purpose of Machine Learning Model	Object Recognition Model, etc.
11.	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration: Cloud Server Configuration :	Local, Cloud Foundry, Kubernetes, etc.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	Technology of Opensource framework
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	e.g. SHA-256, Encryptions, IAM Controls, OWASP etc.
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	Technology used
4.	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	Technology used
5.	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc.	Technology used